

An Ingenious, Piecewise Linear Interpolation Algorithm for Pricing Arithmetic Average Options

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Abstract. Pricing arithmetic average options continues to intrigue researchers in the field of financial engineering. Since there is no analytical solution for this problem until present, developing an efficient numerical algorithm becomes a promising alternative. One of the most famous numerical algorithms for pricing arithmetic average options is introduced by Hull and White [13]. In this paper, motivated by the common idea of reducing the nonlinearity error in the adaptive mesh model [10] and the adaptive quadrature numerical integration method [9], the logarithmically equally-spaced placement rule in the Hull and White's model is replaced by an adaptive placement method, in which the number of representative average prices is proportional to the degree of curvature of the option value as a function of the arithmetic average price. Numerical experiments verify the superior performance of our method in terms of reducing the interpolation error. In fact, it is straightforward to apply this ingenious method to any pricing algorithm with the techniques of augmented state variables and the piece-wise linear interpolation approximation.

Keywords: Arithmetic average options, logarithmically equally-spaced placement, adaptive placement

1 Introduction

Asian Options are path dependent securities whose payoff depends on the average of the underlying prices during the life of the option. They were originally issued in 1987 when Banker's Trust Tokyo office used them for pricing average options on crude oil contracts, and hence the name "Asian" option. The average can be either the arithmetic average or the geometric average.

This type of option contract is commonly traded on currencies and commodity products which have low trading volumes. This is because in situations where the underling asset is thinly traded or there is the potential for its price to be manipulated, an Asian option offers some protection. In addition, Asian options are attractive because they tend to be less expensive than comparable vanilla options, because the volatility in the average value of an underling asset is inclined to be lower than the volatility of the value of the underling asset. In practice, end-users of commodities or energies tend to be exposed to average prices over time, so Asian options are also attractive for them. Asian options are also popular with corporations, such as exporters, who have currency exposures. This is because Asian options are often used as they more closely replicate the requirements of firms exposed to price movements on the underlying asset.

However, the path-dependent feature complicates the evaluation of this type of options. If the stock price process follows the geometric Brownian motion, the analytical derivation of the pricing formula for geometric average options is feasible since the product of lognormally distributed prices remains to follow the lognormal distribution. Based upon this observation, Kemna and Vorst proposed a Black-Sholes-formula-like analytical solution for European geometric average options [14]. Unfortunately, to this date, it is still analytically intractable to price arithmetic average options due to the lack of proper mathematical representation for the sum of lognormal random variables. Therefore, many researches were devoted to deal with the distribution of the sum of lognormal random variables and derive the approximation pricing formula for the arithmetic

average options. Several works along this direction include the fast Fourier transformation in [2], the Edgeworth series expansion in [21], the reciprocal Gamma distribution in [17], the Laplace transform inversion in [12], etc.

The tree-based model is also a possible alternative to derive the value of the arithmetic average options. However, based on the binomial-tree model introduced in [4], the naive pricing method which is able to derive the exact value of the arithmetic average options by recording all possible arithmetic average prices is simply intractable due to the exponential growth of the number of possible arithmetic average prices with respect to the number of time steps. In this paper, the exact option value stands for the option value derived from a pricing method based on the tree-based model without any interpolation error. It is worth noting that the exact option value from the naive pricing method is still susceptible to the discretization error, which is a trait of the tree-based model, but when the size of each time step approaches zero, the exact option value will converge to the true option value.

To overcome the problem of the exponential growth of the number of possible arithmetic average prices, Dai and Lyuu develop a trinomial-tree pricing model for arithmetic average options that guarantees the convergence to the exact option value [7, 8], in which the notion of integrality of stock prices is employed to reduce the time complexity of recording all possible arithmetic average prices. Although the time complexity in their model is proven to be sub-exponential with respect to the number of time steps, it is still intractable to price arithmetic average options via this model when the number of time steps is large.

On the other hand, in Hull and White's model [13], instead of keeping track of all possible arithmetic average prices, representative average prices are logarithmically equally-spaced placed between the maximum and minimum arithmetic average prices for each node. In addition, the piece-wise linear interpolation is adopted to derive the corresponding option values for nonexistent average prices during the phase of backward induction. As a result, the interpolation error occurs and furthermore, the answer to the question whether the interpolation error vanishes is uncertain for all but a single scenario in which the number of representative average prices for each node and the number of time steps for the tree model are well collocated, see [11].

Along with the line of [13], Neave and Turnbull [18] suggest using the conditional frequency distribution to adjust the number of representative average prices for each node. Cho and Lee [3] replace the uniform allocation of the number of representative average prices in the Hull and White's model with the distribution of the number of possible geometric average prices. Klassen [15] proposes a revised version of the algorithm of [13], in which only a set of average prices at each node is considered, and the grid space for the logarithm of the arithmetic average prices is a pre-specified function of the time to maturity, the time steps, and the volatility of the stock price process. Although these methods of adjusting the allocation of the number of the representative average prices over the tree exhibit superior convergence rate to the exact option value than the Hull and White's model, the major disadvantage of them is the absence of the economic meanings and the guarantee of the convergence of the interpolation error.

A different point of view is adopted in [1] and [5, 6] to improve the convergence rate of the tree-based models for pricing arithmetic average options. Instead of recording the maximum and minimum arithmetic average prices, a more compact range is derived such that the interpolation error can be reduced effectively. Moreover, for European-style arithmetic average options, the optimal allocation of the representative average prices over the tree is derived in [5] to minimize the accumulated interpolation error of the option value.

Dedicated to improving the efficiency of allocation the representative average prices over the tree to reduce the interpolation error, the suggested allocation rules of the above modifications are no longer uniformly distributed but are contingent upon several factors, including the probability reaching the underlying node, the time to maturity of the underlying node, the number of time steps in the tree model, and the volatility of the underlying process. The difference between uniform and non-uniform allocations is illustrated in Panel 1 of Figure 1.

With the uniform allocation rule being replaced, the logarithmically equally-spaced placement rule proposed by Hull and White is still retained in the aforementioned works. In this paper, the main contribution is to provide another aspect of minimizing the interpolation error. Aiming at simultaneously guaranteeing the convergence of the interpolation error and improving the efficiency of the Hull and White's model, the logarithmically equally-spaced placement strategy is replaced by an adaptive placement method, in which more representative average values are needed in the area around which the option value as a function of the arithmetic average prices is with higher degree of curvature, and fewer representative average values are placed where the option value function is with lower degree of curvature. The ideas of the adaptive placement

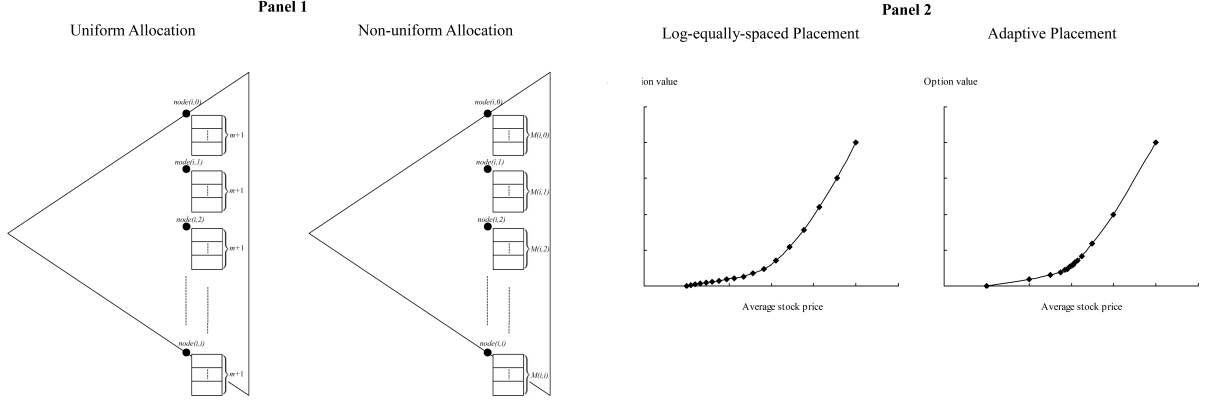


Fig. 1. The illustration of our adaptive placement method. Hull and White [13] adopted the combination of the uniform allocation and logarithmically equally-spaced placement rules in their pricing algorithm, i.e. $m=100$. Other modifications of the Hull and White's model focus on devising more efficient non-uniform allocation rules, i.e. $M(i, j)$ is different for each $node(i, j)$. However, the logarithmically equally-spaced placement rule is a common component in these models. In our adaptive placement method, the number of representative average prices is proportional to the degree of curvature and an efficient non-uniform allocation of representative average prices is achieved automatically.

method as well as the logarithmically equally-spaced rules are illustrated in Panel 2 of Figure 1. To achieve this goal, the adaptive placement method is actually designed to govern the error of the linear interpolation between each pair of adjacent representative average prices under a limit criterion. Moreover, our method forms automatically an efficient non-uniform allocation of the number of representative average prices over the tree.

Extensive numerical experiments are conducted, which not only verify the superior performance of our adaptive placement method in terms of reducing the interpolation error, but also demonstrate that the idea of the algorithms of [1] is incorporated in our adaptive placement method as a natural byproduct. Therefore, we suggest using this novel method as a standard procedure whenever the pricing algorithm employs the techniques of augmented state variables and the piece-wise linear interpolation approximation, for example, the GARCH option pricing algorithm in [20].

This paper is organized as follows. Section 2 describes the arithmetic average options and the famous Hull and White's model and some of its modifications. Our adaptive placement method will be elaborated in Section 3. Numerical results in Section 4 verify the superior performance of our method to existing methods. Section 5 concludes this paper.

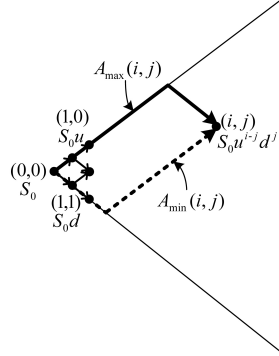
2 Arithmetic Average Options

In this paper, the non-dividend-paying underlying asset price in the risk neutral world is assumed to follow the geometric Brownian motion: $dS_t/S_t = rdt + \sigma dZ$, where r is the risk free rate, σ is the volatility of the asset price, and Z is a Wiener process. Suppose that the stock price is sampled at the time points $0 = t_0 < t_1 < \dots < t_n = T$ during the life of the arithmetic average options. If the corresponding stock prices are $S_{t_0}, S_{t_1}, \dots, S_{t_n}$, the arithmetic average price from time 0 to t is $A(t) = (\sum_{i=0}^l S_{t_i})/(l+1)$, where $t_l \leq t < t_{l+1}$. In addition, the exercise value of the arithmetic average call considered in this paper at time t is $\max(A(t) - X, 0)$, where X is the strike price of the arithmetic average call. Furthermore, the stock price is assumed to be sampled periodically, which is often the case in the real world, and therefore $t_i = i\Delta t$ and $\Delta t = T/n$.

2.1 The Hull and White's Model

In the field of option pricing, the binomial-tree model introduced in [4] is viewed as a useful tool to deal with European-style as well as American-style options. This model divides the time horizon of an option into n

Panel 1. The maximum and minimum average paths



Panel 2. The illustration of the backward induction procedure

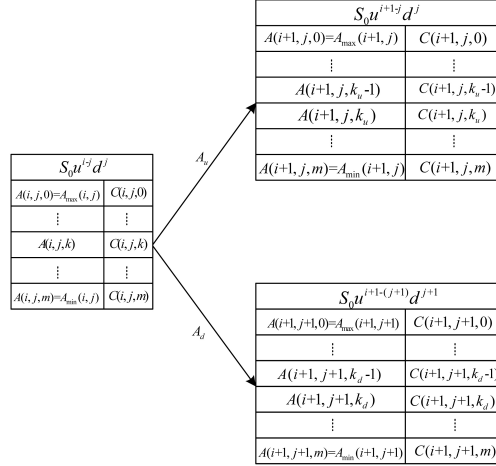


Fig. 2. The paths of maximum and minimum arithmetic average prices and the backward induction procedure in the Hull and White's model. In Panel 1, it is shown that the stock price today is denoted by S_0 , and the stock price can either move up to S_0u or down to S_0d at the first step. Similarly, each stock price can either move up or move down at subsequent time steps. The $node(i, j)$ stands for the node at time point i with j cumulative down movements and the $S_0u^{i-j}d^j$ is the corresponding stock price. $A_{\max}(i, j)$ ($A_{\min}(i, j)$) is the maximum (minimum) average stock price among all possible paths from $node(0, 0)$ to $node(i, j)$. In Panel 2, for each possible average price $A(i, j, k)$, it is necessary to find the corresponding A_u and A_d first and then to derive the option values C_u and C_d by the piece-wise linear interpolation. The continuation value for $A(i, j, k)$ is $C(i, j, k) = (p \cdot C_u + (1 - p) \cdot C_d)e^{-r\Delta t}$.

discrete time steps and discretizes the stock prices at each time step. In Panel 1 of Figure 2, it is shown that the stock price at time step 0 is S_0 (at $node(0, 0)$), and the stock price can either move up to S_0u (at $node(1, 0)$) or down to S_0d (at $node(1, 1)$) at the first time step, where $u = \exp(\sigma\sqrt{\Delta t})$ is the magnitude of a upward movement for the stock price, and $d = \exp(-\sigma\sqrt{\Delta t})$ is the magnitude of a downward movement for the stock price. Similarly, each stock price can either move up or move down at subsequent time steps.

It is in theory possible to employ the binomial-tree model to calculate exact values of arithmetic average options by recording all possible average values reaching each node. Unfortunately, if the option life is divided into n periods, the number of all possible arithmetic average prices is 2^n , which implies that the computation complexity is intractable even for small values of n . In fact, the memory constraint of modern personal computers makes it impossible to deal with the cases when n is larger than 20.

One of the most famous tree-based models to price arithmetic average options efficiently is proposed in [13]. In their algorithm, to avoid tracking all possible arithmetic average prices of each node, only the maximum and the minimum arithmetic average prices of all traversed paths for each node are calculated, which is illustrated in Panel 1 of Figure 2.

For $node(i, j)$ with the stock price $S_0u^{i-j}d^j$ for $0 \leq j \leq i \leq n$, the maximum arithmetic average price is contributed by a price path starting with $i - j$ consecutive up movements followed by j consecutive down movements, whose value can be derived by $A_{\max}(i, j) = (S_0 \frac{1-u^{i-j+1}}{1-u} + S_0u^{i-j}d \frac{1-d^j}{1-d}) / (i + 1)$. Likewise, the value of the corresponding minimum arithmetic average price can be calculated from a price path starting with j consecutive down movements followed by $i - j$ consecutive up movements: $A_{\min}(i, j) = (S_0 \frac{1-d^{j+1}}{1-d} + S_0d^j u \frac{1-u^{i-j}}{1-u}) / (i + 1)$.

Once equipped with the knowledge about the maximum and minimum arithmetic average prices for each node, the logarithmic space between $A_{\max}(i, j)$ and $A_{\min}(i, j)$ is divided into m equal-length sub-intervals and $m + 1$ representative average prices are obtained via $A(i, j, k) = \exp(\frac{m-k}{m} \ln(A_{\max}(i, j)) + \frac{k}{m} \ln(A_{\min}(i, j)))$, for $k = 0, 1, \dots, m$, where k is the index of representative average prices from 0 to m .

After building the tree and the table of representative average prices for each node, we decide the payoff of each representative average price of the nodes at maturity first. Next, the option value is derived via the

backward induction procedure. The backward induction procedure from $node(i+1, j)$ and $node(i+1, j+1)$ to $node(i, j)$ is illustrated in Panel 2 of Figure 2.

For $A(i, j, k)$, the evolutions of the arithmetic average price at the next time point are $A_u = [(i+1)A(i, j, k) + S_0 u^{i+1-j} d^j]/(i+2)$, and $A_d = [(i+1)A(i, j, k) + S_0 u^{i+1-(j+1)} d^{j+1}]/(i+2)$. Suppose that A_u is inside the range $[A(i+1, j, k_u), A(i+1, j, k_u-1)]$. The option value C_u for the arithmetic average price A_u is approximated by the linear interpolation $C_u = w_u C(i+1, j, k_u) + (1-w_u)C(i+1, j, k_u-1)$, where $w_u = (A(i+1, j, k_u-1) - A_u)/(A(i+1, j, k_u-1) - A(i+1, j, k_u))$. Similarly, the option value of C_d for the arithmetic average price A_d is derived from $C_d = w_d C(i+1, j+1, k_d) + (1-w_d)C(i+1, j+1, k_d-1)$, where $w_d = (A(i+1, j+1, k_d-1) - A_d)/(A(i+1, j+1, k_d-1) - A(i+1, j+1, k_d))$, if A_d is inside the range $[A(i+1, j+1, k_d), A(i+1, j+1, k_d-1)]$. As a consequence, the continuation value for $A(i, j, k)$ is

$$C(i, j, k) = (p \cdot C_u + (1-p) \cdot C_d) e^{-r\Delta t}. \quad (1)$$

2.2 Some Modifications for the Hull and White's model

However, the interpolation error is incurred in the Hull and White's model due to the limited number of representative average prices at each node and employing the piece-wise linear interpolation to find option values for nonexistent average prices. The brute-force method via increasing the number of representative average prices for each node is able to enhance the accuracy for the option values of course, but it is accompanied with unacceptable computation time in the meanwhile. Many researches focus on reducing the aforementioned interpolation error in a more efficient way. In this subsection, some modifications of the Hull and White's model are explored. The performance of these modifications as well as the Hull and White's model will be compared to that of our adaptive placement method in the Section 4.

Apply the quadratic interpolation

The quadratic interpolation is actually the second order Lagrange polynomial interpolation method. Based on three pairs of the arithmetic average price and its corresponding option value (A_0, C_0) , (A_1, C_1) , and (A_2, C_2) , the Lagrange polynomial coefficients for an average price A are as follows:

$$L_0 = \frac{(A - A_1)(A - A_2)}{(A_0 - A_1)(A_0 - A_2)}, \quad L_1 = \frac{(A - A_0)(A - A_2)}{(A_1 - A_0)(A_1 - A_2)}, \quad L_2 = \frac{(A - A_0)(A - A_1)}{(A_2 - A_0)(A_2 - A_1)},$$

and the option value C corresponding to A is $C = C_0 \times L_0 + C_1 \times L_1 + C_2 \times L_2$.

Tighten the range for representative average prices

For European fixed-strike-price arithmetic average calls, according to [1], for nodes at time point i , if some average price A is larger than the upper bound $(n+1)X/(i+1)$, because this path is sure to be in the money at maturity, the corresponding expected option value can be calculated directly via

$$e^{-r(n-i)\Delta t} [(i+1)A - (n+1)X + S_0 u^{i-j} d^j e^{r\Delta t} \frac{1 - e^{r(n-i)\Delta t}}{1 - e^{r\Delta t}}] / (n+1). \quad (2)$$

Therefore, the range $[A_{\min}(i, j), A_{\max}(i, j)]$ can be curtailed to

$$[\min(A_{\min}(i, j), (n+1)X/(i+1)), \min(A_{\max}(i, j), (n+1)X/(i+1))]$$

and whenever the average price is above the upper bound, the corresponding expected option value can be derived via Equation (2) without suffering any interpolation error.

3 Our Models

The goal of our adaptive placement method is to intelligently reduce the interpolation error for pricing arithmetic average options in the tree-based model. In this section, the algorithm of our adaptive placement will be elaborated. Motivated by the common idea of dealing with the nonlinearity error in the Figlewski and

Gao's adaptive mesh model [10] and the adaptive quadrature numerical integration method [9], our method differs from the Hull and White's method in the sense that more representative average prices are placed in the range where the option value function is with higher degree of curvature and fewer are placed in the range where the option value function is with lower degree of curvature (see Figure 1).

The details of our adaptive placement method are explained as follows. In fact, the linear interpolation is the first-order special case of the general Lagrange polynomial interpolation, such that for any $A \in [A_1, A_2]$, where A_1 and A_2 stand for any pair of adjacent representative average prices in the table of average prices for some node, the corresponding option value can be derived from $C = C_1 \cdot \frac{A-A_2}{A_1-A_2} + C_2 \cdot \frac{A-A_1}{A_2-A_1}$, where C_1 and C_2 are corresponding option values for A_1 and A_2 . By the mean-value theorem, the error term of the linear interpolation caused from the nonlinearity of the option value function in $[A_1, A_2]$ can be expressed as

$$\frac{C''(\xi)}{2!} \cdot (A - A_1) \cdot (A - A_2), \text{ for some number } \xi \text{ between } A_1 \text{ and } A_2. \quad (3)$$

Our adaptive placement method is designed to examine whether the error of the linear interpolation between each pair of adjacent representative average prices in Equation (3) is below some pre-specified limit. Once the error of the linear interpolation inside the range of $[A_1, A_2]$ is not negligible, i.e. $C''(\xi)$ is too large or the distance between A_1 and A_2 is too far, we divide $[A_1, A_2]$ into finer subsets by inserting an extra representative average price inbetween and then repeat the same procedure of examining the error of the linear interpolation for each subset. Once the value of the error term between any pair of adjacent representative average prices is smaller than the predefined threshold (termed the *second order error criterion* in our method), this examining-and-dividing process is stopped.

In practice, another constant termed the *precision criterion* is also defined to represent the threshold of negligible refinement for both average prices and option values in our method. The above examining-and-dividing process is also terminated when the difference between adjacent representative average prices or their corresponding option values is smaller than this minimum criterion. The purpose of introducing the *precision criterion* is to prevent possibly infinite dividing caused from the non-differentiable point during the examining-and-dividing process. Within each examination of the linear interpolation error, we approximate $C''(\xi)$ in Equation (3) by the second order numerical differentiation. For any pair of adjacent representative average prices A_1 and A_2 , the midpoint $A = (A_1 + A_2)/2$ is employed together to approximate the error term of the linear interpolation for this range.

The steps to price arithmetic average options in this paper are described as follows. During the phase of building the stock price tree, except the uppermost and lowermost nodes at each time step in the binomial tree, only the maximum and minimum average prices for each node are recorded as representative average prices. Meanwhile, we also determine whether the strike price is within the range between the maximum and the minimum average prices. If so, the strike price is inserted into the table of representative average prices. As a consequence, there will be two or three representative average prices for each node after building the stock price tree. Since the number of possible arithmetic average prices for the nodes at the initial three time steps is not larger than three, it is not necessary to decide representative average prices via the above procedure and instead we record all possible average prices for these nodes.

After building the stock price tree, the tables of representative average prices for all nodes are mainly constructed during the phase of backward induction, which is the most different part from existing models for pricing arithmetic average options. For each node, the goal of adaptively placing representative average prices is to let the range between each pair of adjacent representative average prices be small enough such that the linear interpolation error throughout the range is smaller than the *second order error criterion*.

We take an example to illustrate the examining-and-dividing process of our adaptive placement method step by step. Suppose $S_0 = X = 50$, $n = 40$, $T = 1$, $r = 10\%$, $\sigma = 80\%$, and both the *second order error criterion* and the *precision criterion* are 0.5. For *node(37, 25)*, the examining-and-dividing process is sketched in Figure 3. Inside the frame of each step, there are three pairs of representative average prices and the corresponding call values, and we also report the linear interpolation error when these three pairs of representative average prices and option values are considered.

When the backward induction progresses to *node(37, 25)*, only 83.4062, 50, and 12.3309 are in the initial table of representative average prices, and according to the Hull and White's model, their cor-

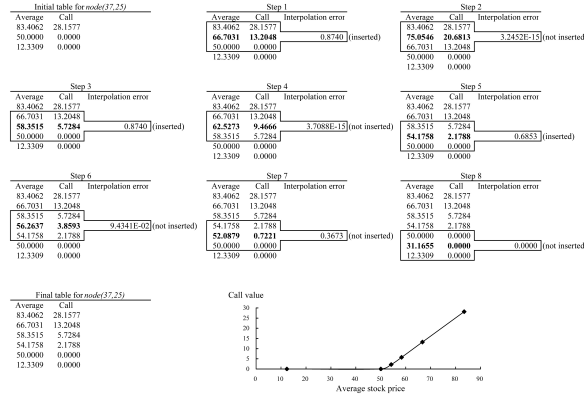


Fig. 3. The numerical example for the examining-and-dividing process. The values of parameters in this example are $S_0 = X = 50$, $n = 40$, $T = 1$, $r = 10\%$, $\sigma = 80\%$. In addition, the *second order error criterion* and the *precision criterion* are both 0.5. These figures illustrate the examining-and-dividing process for *node(37, 25)*. Once the approximate linear interpolation error is larger than 0.5, the pair of the average price and the call value in boldface will be inserted into the table of representative average prices of *node(37, 25)*. The approximate linear interpolation error for any pair of adjacent representative prices in the final table is bounded by the *second order error criterion*.

responding option values are 28.1577, 0, and 0 respectively.⁴ In step 1, the pairs of (83.4062, 28.1577), ((83.4062+50)/2=66.7031, 13.2048), and (50, 0) are considered to approximate the linear interpolation error for the range between 83.4062 and 50. Because the approximate linear interpolation error 0.8740 is larger than the *second order error criterion* 0.5, the pair of the average price 66.7031 and the corresponding call value 13.2048 should be inserted into the table of representative average prices. In step 2, the approximated linear interpolation error in the range between 83.4062 and 66.7031 is 3.2452E-15, which is smaller than the *second order error criterion* 0.5. Therefore, we do not insert the pair of the average price 75.0546(=(83.4062+66.7031)/2) and the corresponding option value 20.6813 since the linear interpolation works pretty well in the range between 83.4062 and 66.7031. Following the same reasoning, we can derive the final table of representative average prices and their corresponding option values through steps 3 to 8, in which the approximate linear interpolation error for any pair of adjacent representative average prices is below the *second order error criterion* 0.5.

In addition, the option value as the function of the average price is plotted in Figure 3. It is apparent that the above examining-and-dividing process will balance the distribution of placed representative average prices so that it becomes proportional to the degree of the curvature of the option value as a function of the arithmetic average price. Since the performance of the piece-wise linear interpolation is poor around where the option value function is with high degree of curvature, our algorithm places more representative average prices in these areas to reduce the error of the piece-wise linear interpolation. On the other hand, due to the satisfactory performance of the piece-wise linear interpolation for dealing with the option value function with low degree of curvature, our algorithm argues that less representative average prices placed in those areas will be sufficient.

4 Numerical Results

4.1 Comparisons with the Hull and White's Model

It is well known that representative average prices are logarithmically equally-spaced placed in the Hull and White model [13], whereas the number of placed representative average prices is proportional to the degree of the curvature of the option value function in our adaptive placement method. The differences are shown in Figure 4.

⁴ We derive the corresponding option values following the Hull and White's algorithm, i.e. via Equations (1) in Section 2.1.

Panel 1. The distribution of representative average prices at maturity (node(40, 20))

Panel 2. The distribution of representative average prices at node(34, 13) (European arithmetic average call)

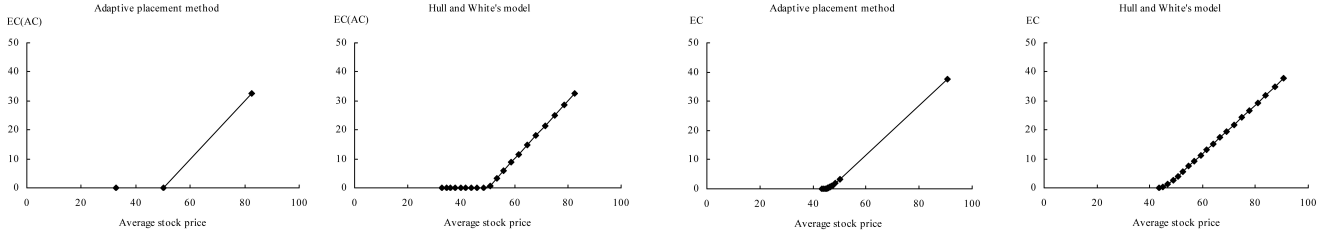


Fig. 4. Comparisons of the distributions of representative average prices of the adaptive placement method and the Hull and White's model. For the readability of this figure, the values of parameters are specified as: $S_0 = X = 50$, $n = 40$, $T = 1$, $r = 10\%$, $\sigma = 30\%$, the *second order error criterion* is 0.01, and the *precision criterion* is 0.001. In addition, the number of representative average prices in the Hull and White's model is 20. It is apparent that our adaptive placement method places more representative average prices in the region with higher degree of curvature, but the Hull and White's model always places representative average prices logarithmically equally-spaced between the maximum and minimum average prices.

In Panel 1 of Figure 4, if the strike price is between the maximum and minimum arithmetic average prices for some node at maturity, it is easily found that there is no linear interpolation error for both linear segments, and therefore it is not necessary to insert any representative average price. However, the Hull and White's model still employs $m + 1$ representative average prices for each node at maturity. For each linear segment, our method provides the interpolated results as accurate as those in the Hull and White's model, but near the kink, our method will outperform the Hull and White's model except the strike price happens to be one of the representative average prices in the Hull and White's model.

In Panel 2 of Figure 4, it is clear that the logarithmically equally-spaced placement in the Hull and White's model places too many representative average prices on the region with low degree of curvature, but only a few representative average prices are needed in our adaptive placement method to derive interpolated results with sufficient accuracy in this region. On the contrary, to deal with regions with high degree of curvature, the Hull and White's model is inclined to generate unexpected large pricing errors, since it does not consider the nonlinearity error.

4.2 Convergence Rates with respect to the Number of Representative Average Prices

This section compares the rate of convergence with respect to the number of representative average prices for different methods. The values of parameters in our numerical example are as follows: the stock price S_0 is 50, the strike price X is 50, the time to maturity T is 1 year, the risk-free interest rate r is 10% per annum, the volatility σ is 80%, the time step n is 40, and different numbers of representative average prices are examined. In order to obtain a better understanding of the rates of convergence, an analysis on the relative error and the number of representative average prices is performed. Plots of $\ln(|\text{relative error}|)$ in relation to the number of representative average prices for the arithmetic average calls are in Figure 6. In the figure, a steeper slope represents a better convergence rate, and the lower the line is located, the less relative error that the underlying method is generated under the same number of representative average prices.

It is apparent the Hull and White's model results in the poorest convergence rate, but the relative error decreases significantly when the strike price X is inserted as a representative average price. This is because for nodes near maturity, the kink is near where the average price is equal to the strike price, and the piece-wise linear interpolation is inclined to overestimate the option value around the kink. As a result, the continuation value derived via the Hull and White's model is overestimated significantly.

Finally, it is obviously found that the improvement is minor when combining the AMO algorithms [1] with our adaptive placement method. The reason is analyzed as follows. The idea of the AMO algorithm is to derive the option values of the arithmetic average prices higher than some threshold without incurring any interpolation error and meanwhile concentrate the computation resource (the number of representative average prices) on a smaller range to further reduce the interpolation error. However, for the region above

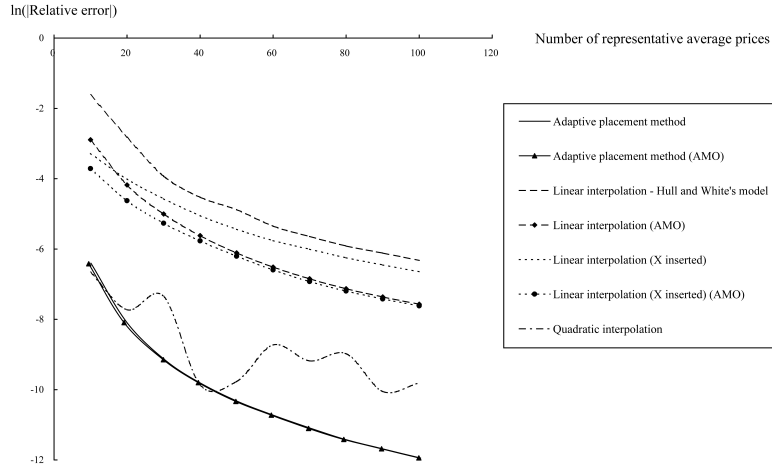


Fig. 5. The rates of convergence of seven different methods of pricing arithmetic average calls. In addition to the results of our adaptive placement method, the results of linear and quadratic interpolations are reported as well. The effect of inserting the strike price as a representative average price is also examined. It is obvious that our adaptive placement method converges faster than other methods with respect to the number of representative average prices. The other observation is that except our adaptive placement method, the convergence rates of other methods are improved when the algorithm of AMO is applied.

the threshold, the interpolation error is in fact very small, so our adaptive placement method already places fewer representative average prices in the region above the threshold, and automatically concentrates on dealing with the region below the threshold. It is believed the concept of the AMO algorithm is already nested in our adaptive placement method.

5 Conclusion

In this paper, the adaptive placement method is proposed to price the arithmetic average options, in which the error of the linear interpolation between each pair of adjacent representative average prices is examined such that representative average prices are placed effectively to reduce the interpolation error. In addition, some modifications of the Hull and White model are also considered, including inserting the strike price into the average price table, the AMO algorithms in [1], and replacing the linear interpolation with the quadratic interpolation.

From the numerical experiments, our adaptive placement method shows great superiority of reducing the interpolation error over other methods. In addition, due to the fact that introducing the AMO algorithms into the adaptive placement method does not improve the convergence rate, we believe that the adaptive placement method already incorporates the idea of this algorithm implicitly.

Since the adaptive placement method is good at reducing the interpolation error efficiently, it is proposed that this technique can be applied to the Hull and White's model as well as any tree-based model with augmented state variables and the piece-wise linear interpolation approximation.

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